

Polychlorinated naphthalenes in human adipose tissue from New York, USA.

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Polychlorinated naphthalenes (PCNs) are persistent, bioaccumulative, and toxic contaminants. Prior to this study, the occurrence of PCNs in human adipose tissues from the USA has not been analyzed. Here, we have measured concentrations of PCNs in human adipose tissue samples collected in New York City during 2003-2005. Concentrations of PCNs were in the range of 61-2500pg/g lipid wt. in males and 21-910pg/g lipid wt. in females. PCN congeners 52/60 (1,2,3,5,7/1,2,4,6,7) and 66/67 (1,2,3,4,6,7/1,2,3,5,6,7) were predominant, collectively accounting for 66% of the total PCN concentrations. Concentrations of PCNs in human adipose tissues were 2-3 orders of magnitude lower than the previously reported concentrations of polychlorinated biphenyls (PCBs) and polybrominated diphenyl ethers (PBDEs). Concentrations of PCNs were not correlated with PCB concentrations. The contribution of PCNs to dioxin-like toxic equivalents (TEQs) in human adipose tissues was estimated to be <1% of the polychlorinated dibenzo-p-dioxin/dibenzofuran (PCDD/F)-TEQs.